Au-delà de nos limites biologiques (Radman with Carton, 2011)

Here's a detailed English summary of *Au-delà de nos limites biologiques* (Radman with Carton, 2011), organized around the scientific core.

Central thesis: aging is protein corrosion

Radman's organizing claim is that aging is fundamentally **oxidative damage to proteins** — what he calls "corrosion" or "rust" of the biological machinery — rather than primarily damage to DNA. The reasoning: DNA is the blueprint (the architect), but proteins are the workers that execute every cellular function, including the repair and maintenance of DNA itself. Even an intact genome produces a failing organism if its proteins degrade.

He marshals several lines of evidence against the DNA-mutation theory of aging:

- Somatic mutations increase only slightly (often negligibly, and roughly linearly) with age, whereas morbidity and mortality rise **exponentially** (Gompertz law).
- Mutations are too rare and scattered to plausibly hit the same genes across most cells of an organ.
- Earl Stadtman showed (from 1988) that protein oxidation rises exponentially with age, mirroring the mortality curve — but it was unclear whether this was cause or consequence.
- Radman's own lab work (with Anita Krisko, Magali Leroy, Meselson) argued that oxidation is the **cause**: decline of cellular function and cell death correlate with protein oxidation, not with DNA damage.

A key mechanistic detail relevant to your work: a **single amino-acid difference** (within normal human polymorphism) can make a protein ~10× more vulnerable to oxidation. So each person's "silent mutations" subtly tune which proteins burn fastest, and the most vital among those burning proteins determines which age-related disease strikes first and when.

The common-cause argument

Because all age-related diseases rise with the same exponential kinetics, Radman proposes they share a **common cause** — protein oxidation. If true, intervening on that single upstream cause would push back the whole spectrum of age-related disease simultaneously, rather than fighting each disease individually. This reframes the goal as **prevention by protection** rather than therapy by repair.

Robust organisms as the model system

The conceptual move that drives the book: classical molecular biology studies *loss-of-function* mutations in normal ("wild-type") organisms, which teaches what breaks function but nothing

about how to *exceed* normal performance. To learn how to improve on the human baseline, you study natural "super-performers" — robust and near-immortal organisms:

- **Deinococcus radiodurans** ("Conan the Bacterium") — discovered in 1956 contaminating gamma-irradiated corned beef; survives ~1 million rads (humans die at ~100s of rads), plus extreme desiccation. Genome sequenced by Venter's institute.
- **Bdelloid rotifers** — ~1,000 cells, 16 neurons; asexual.
- **Tardigrades** — ~40,000 cells, with nervous system and gut.
- Resurrection plants (rose of Jericho), ancient viable seeds, the immortal jellyfish *Turritopsis nutricula*.

Two mechanistic findings from Radman's lab

1. DNA reconstitution (Zahradka, Slade; *Nature*, Oct 2006). *Deinococcus* is constitutively polyploid (always ≥2 identical genome copies, 6–8 when growing fast). When radiation shatters both copies into hundreds of fragments — but at *different* break points — the cell reassembles a complete genome by overlap-matching the fragments, like reconstructing a book from two differently-shredded identical copies. The analogy he uses is the shotgun-sequencing assembly algorithm, except *Deinococcus* does it physically in about an hour. Crucially, **the repair enzymes are essentially the same as in sensitive species** — robustness isn't from novel repair machinery.

2. Protection, not repair, is the secret (Krisko, with Daly's prior comparative work). The robust organisms instead evolved a **non-enzymatic, small-molecule system that protects proteins from oxidation** — especially protecting the repair proteins themselves. Daly developed the concept by comparing bacterial species; Radman's group established the cause-effect link. A **cocktail of small molecules extracted from *Deinococcus* protected the proteins of a radiation-sensitive species.** These protectors are low-molecular-weight (not proteins, not DNA), apparently non-toxic, and — the universalist claim — **protect across species**, including potentially human proteins. Krisko/Meselson (PNAS, late 2010) reported that diverse organisms are equally *sensitive* to protein oxidation; the robust ones simply *protect* their proteins better.

The oxidation chemistry

Radman explains free radicals as oxygen species with unpaired electrons that are highly reactive and propagate damage in chain reactions (his "frustrated singles / badly-married couples" metaphor). ~99% of radicals resolve harmlessly to water; the ~1% remainder does the damage. Three interconverting reactive species (superoxide, hydrogen peroxide, hydroxyl) make the chemistry a network, which is why single-point antioxidants (vitamin C, vitamin E, cosmetic "anti-oxidant" products) fail — you must act on the **whole network at once**, not block one node. This is his explicit critique of conventional antioxidant cosmetics and supplements, directly relevant to your dermocosmetic framing.

Cells already contain radical traps (glutathione, cysteine, cystamine) and detox enzymes (catalase, superoxide dismutase), but these are themselves oxidizable and insufficient. The lab-bench analogy he uses is β-mercaptoethanol acting as a sacrificial reductant protecting enzymes

in storage. The strategy: install more "oxygen traps" so radicals attack expendable molecules instead of vital proteins and DNA.

He also frames the cellular maintenance system as "molecular emergency physicians": **diagnosers** (detect damage), **surgeons/therapists** (enzymes that excise and replace damaged segments — works for DNA, but oxidized proteins are unrecoverable and must be scrapped and resynthesized), and **cleaners** (remove oxidized waste). All are proteins, hence all corrodible — so protecting the protectors is the leverage point.

Aging-relevant biology covered

- **Telomeres/telomerase** (Blackburn, Greider, Szostak — 2009 Nobel): telomere shortening halts division; cancer reactivates telomerase; late-2010 work showed the fate is reversible by artificial telomerase activation. Radman notes this doesn't explain aging in non-dividing cells (neurons, cardiac myocytes), which can't age via telomere loss.
- **Sirtuins, klotho, mitochondria** as other "star" players — but he criticizes researchers for selling their particular mechanism as *the* cause of aging (his "you can stop a car with a club, but the club isn't the usual cause of breakdowns" critique).
- **Caloric restriction**: ~30% lifespan extension in mice; mutations mimicking it give the same; simpler animals can double lifespan.
- **The "babies are born young from old parents" puzzle** (posed by Antoine Danchin): newly synthesized embryonic proteins are functionally young; as long as the proteins that make proteins (including repair/maintenance proteins) stay high-quality, you keep making good proteins and stay young.

Diagnostic application: "bio-gerontometry" / biology of destiny

With Anita Krisko, Radman proposes measuring each individual's **protein oxidation level** as a quantitative readout of true **biological age** (vs. chronological age) — a "biological clock" measurable from a tiny tissue/blood sample via an automated machine that lyses cells, extracts proteins, and runs a specific oxidation-quantifying reaction. Protein-specific measurement could be predictive: e.g., if p53 is more oxidized in one person, higher cancer predisposition; if alpha-synuclein oxidizes faster, earlier Parkinson risk. The aim is individualized **preventive** medicine — not changing one's genetic destiny but *delaying* it in time. He flags the ethical hazard (employers/insurers demanding biological-age data).

Why aging looks complex but may be simple

Radman argues a simple *cause* can have complex *consequences* — even identical mass-produced cars fail differently. So the complexity of aging's manifestations says nothing about the complexity of its cause. He's sharply critical of how the grant/funding system ("corporatist culture") rewards conformist, short-term, application-oriented work and rejects simple or ambitious hypotheses, suppressing the fundamental research that produces unpredictable breakthroughs.

Evolutionary and speculative material

- **"Nécro-sexe" hypothesis** (with Meselson): robust asexual organisms have mosaic genomes with genes "stolen" from all kingdoms (e.g., plant **LEA genes** for desiccation resistance, present in both *Deinococcus* and rotifers). The speculative scenario: during droughts, survivors rehydrate and take up DNA from the dead, integrating useful robustness genes; having painstakingly assembled an optimal genome, they abandon sex to avoid diluting it. (He flags this as plausible-but-unproven and notably anthropomorphized.)
- Only **asexual** rotifers (bdelloids) are robust; sexual ones are as radiation-sensitive as anything else. He notes human neurons and cardiomyocytes resemble asexual rotifer cells in not dividing.
- **HGM (genetically modified humans)**: his provocation that humans are already "GMOs" through evolution; he discusses heritable cancer resistance via extra tumor-suppressor gene copies (the trisomy-21 / reduced-cancer observation with Jean-Claude Weill), and the CCR5-type HIV-resistance mutation as a model for transferable protective alleles.
- Cryonics/suspended animation, frozen frogs, antifreeze proteins and trehalose, the Hayflick limit, HeLa cells, and broad social/ethical speculation (longevity and inequality, work, family, the "ant colony" specialization-vs-diversity tension).

Most relevant threads for dermocosmetics

For your NAOS/Radman mining, the concentrated payload is: **(1)** protein carbonylation/oxidation as the master clock; **(2)** the failure of single-target antioxidants vs. the need for network-level protection; **(3)** the small-molecule, non-enzymatic, cross-species-transferable protectant cocktail from extremophiles; **(4)** protecting the maintenance/repair machinery itself as the leverage point; **(5)** the explicit skin examples — age spots as visible protein/lipid oxidation, Stadtman's exponential protein oxidation in human *skin* cells, and his direct critique of cosmetic antioxidant marketing.